FinInteract: A Bilingual Benchmark for Evaluating Financial Search Agents on Ambiguous Queries

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ABSTRACT

Financial question answering benchmarks assume that every query is fully specified, yet real analyst queries routinely carry unresolved ambiguity: the same question—*"What was Meta Platforms' operating income?"*—can yield answers differing by 34% depending on whether the scope is consolidated total or the Family of Apps segment. We introduce FinInteract, a bilingual (English/Chinese) benchmark of 173 instances that pairs each question Q with a *default interpretation* (the answer a non-expert would assume) and an *intended interpretation* (uniquely identified by a disambiguating context C). Instances span five axes of financial ambiguity—temporal scope, metric definition, entity scope, filing vintage, and recognition policy—drawn from SEC EDGAR filings and A-share annual reports. We evaluate frontier models under three ReAct-based modes (Answer-only, Answer+Search, Answer+Search+Interact), reporting standard *accuracy* alongside interaction-quality diagnostics (interaction rate, axis-targeting precision, calibration). A controlled re-grading shows that conventional single-gold scoring—which credits the naive default reading—over-states model competence by up to 3×, exactly the blind spot FinInteract is built to expose. Even allowed to ask clarifying questions, frontier agents leave 92% of ambiguous queries unresolved (best model 20%; 72% are solved by no model in any mode), yet when handed the resolved interpretation the same models answer 93–95% correctly—so the ≈75-point gap is the cost of failing to *elicit* the disambiguation, not missing knowledge. Forcing models to ask more does not close it (and often hurts), and the capability does not improve monotonically with model scale.

CCS CONCEPTS

• **Information systems** → **Question answering**; • **Computing methodologies** → *Natural language processing*.

KEYWORDS

financial question answering, ambiguity resolution, interactive agents, benchmark, disambiguation, large language models

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1 INTRODUCTION

Large language models are increasingly deployed in high-stakes financial workflows: earnings-call analysis, SEC filing review, analyst report generation, and investor query handling. Yet the benchmarks used to evaluate these models share a critical blind spot—they assume that every query is fully specified before the model begins to answer.

Consider the question *"What was Meta Platforms' operating income?"* This question is genuinely ambiguous: should *operating income* be the consolidated GAAP figure for the total company ($46.75B for FY2023), or the Family of Apps segment figure ($62.87B)? Both are correct under their respective interpretations; the answers differ by 34%. A skilled financial analyst would not guess—they would ask a clarifying question. Existing benchmarks reward the model that guesses correctly; they do not distinguish a model that recognizes the ambiguity from one that happens to pick the default interpretation.

Table 1: Running example: a FinInteract instance. The same question yields two defensible answers. Without the disambiguating context C, a model's default interpretation leads to the wrong answer.

Field	Value
Q	What was Meta Platforms' operating income?
C	Total company (consolidated) results under U.S. GAAP for the year ended December 31, 2023.
A (intended)	$46.75B *[consolidated GAAP]*
A_d (default)	$62.87B *[Family of Apps segment]*
Ambiguity axis	Entity scope
H_0	1.58 bits
Intended span	"Income from operations for 2023 was $46.75 billion..."
Default span	"Family of Apps...Income from operations $62,871..."

Limitations of Existing Benchmarks. FinanceBench [4] provides 150 expert-curated questions over SEC filings, each with a unique correct answer verified against the source document; queries are pre-disambiguated by construction. FinQA [2] and DocFinQA [10] target numerical reasoning over earnings reports but give the model the exact evidence table alongside the question, bypassing retrieval and interaction entirely. BizBench [5] and FinBen [13] aggregate multiple financial tasks but treat QA as closed-book or retrieval-augmented reading comprehension. None of these benchmarks ask whether a model *recognizes* that a question is ambiguous, nor whether it can *resolve* that ambiguity through strategic interaction with a user.

The Ambiguous-QA Gap in Finance. The ambiguous-QA literature has matured in the general domain: AmbigQA [9] introduced multi-answer formulations; CondAmbigQA [7] added condition-structured annotations; CLAM [6] demonstrated that clarification pipelines improve accuracy on ambiguous queries. However, all of this work operates on general-domain Wikipedia questions. Finance is structurally different in three ways. First, financial ambiguity is *endemic and systematic*—a single number such as "net income" can legitimately differ by 30–40% depending on reporting scope, accounting basis, and filing vintage. Second, *misanswers are costly*—an investor acting on the wrong revenue figure or a compliance officer citing the wrong filing version can cause real financial harm. Third, *ground truth is programmatically verifiable*—SEC filings and XBRL facts provide exact machine-readable values for every metric, period, and scope.

The Interaction Stagnation Hypothesis. InteractComp [1] established an important longitudinal finding: while retrieval accuracy on web search benchmarks has improved rapidly with model scale, interaction capability has not kept pace. Across eight frontier models on general-domain tasks, fewer than 14% of questions were answered correctly in full-interaction mode, even though forced-interaction experiments showed the same models could achieve 60–70% accuracy when required to elicit disambiguating context. We hypothesize that this stagnation is more severe in finance, where *domain knowledge is required to even recognize the axes of ambiguity*: a model that does not know what "non-GAAP" means cannot recognize that "adjusted net income" is under-specified.

Our Proposal: FinInteract. We adapt the InteractComp methodology to the finance domain, introduce a five-axis financial ambiguity taxonomy, and release FinInteract —a bilingual benchmark of 173 instances (53 English / 120 Chinese) designed to measure whether financial search agents can recognize query ambiguity and resolve it through efficient interaction.

Contributions:

(1) **Benchmark**: 173 bilingual instances with paired default/intended interpretations, both grounded in verbatim filing quotes, verified by adversarial LLM trials and human annotation (§4).
(2) **Error Taxonomy**: A seven-type error decomposition framework (E1–E7) that maps observed evaluation signals to actionable failure modes (ambiguity blindness, wrong-axis clarification, retrieval dominance, grounding failure) and links them to existing metrics (§7).
(3) **Method Baseline**: Axis-Aware Clarification ReAct —a three-component intervention (ambiguity pre-check, axis-conditioned clarification, interpretation-state-gated answering) that directly targets E1–E3 errors and is evaluated against standard ReAct and oracle baselines (§7).

2 TASK DEFINITION

2.1 Problem Formulation

Each FinInteract instance is a tuple $(Q, C, A, A_d, \mathcal{I}_{\text{int}}, \mathcal{I}_{\text{def}})$:

- Q is an ambiguous financial question over a public company filing.
- C is a disambiguating context: the minimal set of scope or definitional constraints that uniquely collapses Q to A.
- A is the correct answer under the *intended* interpretation $\mathcal{I}_{\text{int}}$.
- A_d is the correct answer under the *default* interpretation $\mathcal{I}_{\text{def}}$ (the answer a non-expert reading Q alone would assume), with $A_d \neq A$.
- $\mathcal{I}_{\text{int}}$, $\mathcal{I}_{\text{def}}$ are structured records specifying entity scope, period, metric, and reporting basis.

An agent reading Q alone (without C) must find it genuinely ambiguous—both A and A_d are defensible from publicly available filings. This *default-vs-intended pairing* generalizes InteractComp's target-distractor design to the financial domain.

2.2 Design Principles

Easy to verify, Ambiguous to resolve. Adopted from InteractComp: ground truth is programmatically verifiable against SEC XBRL facts and CNINFO-structured data, yet the question cannot be correctly resolved without the disambiguating context C.

No yes/no questions. Q must elicit a substantive financial value, not a binary confirmation.

Company must be named in Q. The question is anchored to a specific issuer (English: capitalized proper noun; Chinese: CJK company name).

Context must not contain the answer. C specifies scope constraints but not the numeric value A itself.

2.3 Agent Interaction Model

We follow InteractComp's ReAct framework [14]. The agent may issue three types of actions: **search**(q)—retrieves a passage from the filing corpus; **interact**(q)—presents a yes/no question to a simulated user who answers based on C; **answer**(v)—submits a final answer. We evaluate agents in three primary modes (Table 2) and four ablation modes.

Table 2: Evaluation modes.

Mode	Actions	Description
Answer-only	answer	Pure parametric recall
Answer+Search	search, answer	Oracle retrieval augmented
Answer+Search+Interact	search, interact, answer	Standard full interaction
Axis-Aware ReAct	state, search, interact, answer	**Proposed: axis-conditione**
Always-ask	Must ask ≥ 1 question before answering	
Axis-oracle	Told the primary axis, not the value	
Template-oracle	Fixed human-written question per axis	
Enumerate	Lists all interpretations and answers	

3 FINANCIAL AMBIGUITY TAXONOMY

We define five axes of financial ambiguity, each grounded in standard accounting and disclosure practice (Table 3).

Table 3: Five-axis financial ambiguity taxonomy.

Axis	Definition
Temporal scope	Ambiguity over which time period is intended: fiscal year ending September vs. calendar year; Q3 results vs. full year; TTM vs. point-in-time.
Metric definition	Ambiguity over accounting basis: GAAP net income vs. non-GAAP adjusted income; organic revenue vs. as-reported; operating EBITDA vs. net income.
Entity scope	Ambiguity over consolidation level: segment vs. consolidated total; parent-attributable vs. full consolidated; Class A vs. Class B shares; A-shares vs. H-shares.
Filing vintage	Ambiguity over filing version: original 10-K vs. amended 10-K/A; as-reported vs. restated; preliminary earnings release vs. final filing.
Recognition policy	Ambiguity over accounting policy: revenue recognized point-in-time vs. over time; gross vs. net revenue; capitalized vs. expensed R&D.

Each instance exercises one primary axis (mandatory) and optionally one or two secondary axes where the passage genuinely supports additional ambiguity. Viable structured ambiguity is far more available for some axes than others: cleanly paired, independently verifiable ambiguity is abundant for entity scope and metric definition but rare for the other three in public filings (Appendix A). The released corpus reflects this—entity scope (54%) and metric definition (37%) dominate, with recognition policy (5%), temporal scope (4%), and filing vintage (0%) sparse. We therefore position FinInteract as an **entity/metric-dominant** financial-ambiguity benchmark with exploratory coverage of three sparser axes, confine quantitative per-axis claims to the two well-powered axes, and report the sparser axes with their sample sizes as preliminary signal.

Disambiguation Entropy. For each instance we compute the prior disambiguation entropy: $H_0 = \log_2 k$, where k is the number of plausible interpretations before any clarifying interaction. For a single-axis instance with two interpretations, $H_0 = 1$ bit. For a two-axis instance with three and four plausible values respectively, $H_0 = \log_2(3\times4) \approx 3.58$ bits. This quantity anchors our supplementary efficiency metric DisE$^+$ (§5.3) and serves as a per-instance difficulty proxy.

4 DATASET CONSTRUCTION

4.1 Data Sources

English (EN, ~80% of pool). *DocFinQA*—780 long-context passages extracted from 10-K and 10-Q filings, with verified candidate answers from annotated QA pairs. *EDGAR (XBRL-derived)*—111 passages built from machine-readable XBRL facts for S&P 500 and Russell 1000 companies (FY2024–2025). Because instances are newly constructed from structured facts requiring fresh ambiguity pairs, memorization risk is substantially reduced relative to benchmarks that reuse published QA pairs directly—though we do not claim absolute contamination safety.

Chinese (ZH, ~20% of pool). *CNINFO/akshare*—237 passages derived from A-share annual reports (上证 50 + 沪深 300 constituents) via structured financial statement APIs. Three passage types: (1) metric definition—扣非净利润 vs. 归母净利润 (non-recurring-excluded profit vs. parent-attributable profit); (2) entity scope—归母净利润 vs. 净利润 (parent-attributable vs. fully consolidated including minority interest); (3) temporal scope—year-over-year revenue growth. ZH instances are expected to be harder for current models due to sparser training data on A-share filings and more complex corporate structure nomenclature (e.g., A/H-share distinctions, 集团 vs. 子公司 scope).

4.2 Three-Role Construction Pipeline

Instance construction follows a three-role pipeline. Each passage attempt costs approximately 10–14 minutes of wall time, dominated by two sequential GPT-5 calls.

Step 1: Constructor (GPT-5, ~2–5 min). A single GPT-5 call with a 4,096-token budget generates the full (Q, C, A) triple given a filing passage, verified candidate answer, and mandatory primary ambiguity axis. The output JSON includes `question`, `context`, `answer`, `default_answer`, `intended_evidence_span`, `default_evidence_span`, `intended_interpretation`, `default_interpretation`, and `axes_exercised`. Axis-specific instructions are injected per primary axis to prevent the constructor from drifting toward whichever axis is easiest to generate.

Step 2: Pre-verifier sanity checks (~instant). Thirteen automated code-level rules are applied without any API calls (Table 4). Any failure immediately rejects the passage, avoiding wasted verifier budget.

Step 3: Adversarial Verifier (10 trials, parallelized, ~5–10 min). Ten verifier calls fire simultaneously via a thread pool: five GPT-5-mini trials and five GPT-5 trials. Each trial asks the model to answer Q *without* context C and to state the interpretation it assumed (period, entity, metric, basis). An instance is **rejected** if ≥ 2 of the 10 trials both (a) produce the correct answer A and (b) state an assumed interpretation that aligns with the intended interpretation. This two-condition criterion prevents false rejections from lucky guesses: if a model produces the right number while assuming a different interpretation (e.g., the default), the coincidence is not counted against the instance. Because all ten calls run concurrently, wall time equals the latency of the slowest single GPT-5 call.

4.3 Axis Diversity Enforcement

A naive construction loop over-represents axes common in the passage pool. We enforce target axis shares through four mechanisms: (1) a priority function with a 3× penalty on over-quota axes; (2) hard exclusion at 2× quota; (3) rarity-adjusted initial sort (passages sorted by $\sum_{\text{axes}}(\text{target\_share}/\text{pool\_frequency})$); and (4) dynamic re-sort every 10 instances.

Table 4: Pre-verifier sanity checks (14 rules).

Rule	Description
R1	Q must not be answerable yes/no
R2	Q must not contain disambiguating terms or inline dates
R3	A taken verbatim from the source candidate value (not invented)
R4	Q must name the company
R5	Q must ask about a substantive financial metric
R6	Answer type consistent with question type
R7	Context C must not contain the answer value
R8	Intended and default interpretations must differ
R9	All axis labels must be valid vocabulary items
R10	Answer must be non-trivial (not empty or "0")
R11	`default_answer` present and differs from `answer`
R12	Both evidence spans non-empty
R13	Evidence spans must be meaningfully distinct (≤85% overlap)
R14	Intended and default *answers* differ beyond grader tolerance

4.4 Human Validation Protocol

Two finance-literate annotators (one bilingual, for the Chinese split) independently review instances on an eight-question protocol (H1–H8): H1 *Is Q ambiguous without C?* H2 *Is the default interpretation plausible for a non-expert?* H3 *Does C uniquely identify the intended interpretation?* H4 *Does C avoid directly stating the answer?* H5 *Is A correct under the intended interpretation?* H6 *Is A_d correct under the default interpretation?* H7 *Is the primary axis label correct?* H8 *What yes/no question would a financial analyst naturally ask to resolve the ambiguity?* (free text, used to compute human AxisHit). Acceptance requires H1–H6 to hold after adjudication by the primary author.

Agreement. On a stratified 60-instance sample (by language and axis), per-item raw agreement is 0.82–0.97. Because the labels are heavily skewed toward "yes" (> 85% per item), Cohen's κ is degenerate under the Feinstein–Cicchetti prevalence paradox; we therefore report the prevalence-robust Gwet's AC1, which is 0.85 **on average** (range 0.78–0.97) and 1.00 for the axis label (H7)—substantial-to-near-perfect agreement. Only one instance was rejected by *both* annotators on the same item (consensus rejection 1.7%); adjudication confirmed it as an axis over-reach (a revenue-recognition-timing pairing whose plausible default is the firm's total revenue, not a disaggregation component). We quarantine it from the public release (`v1.1`, N=172); all experiments below are computed on the frozen `v1.0` evaluation snapshot (N=173), which retains it. The instance is unsolved by every model in every *interactive* mode—it is answerable only when the resolving context is supplied outright (context-oracle), where removing it shifts the ceiling by < 0.05pp (93–95% unchanged)—so no reported finding depends on it.

4.5 Dataset Statistics

The frozen evaluation snapshot (`v1.0`) contains 173 instances (Table 5; the public `v1.1` release is N=172, §4.4), drawn from 61 unique companies and SEC/CSRC filings dated 2022–2025. Every instance passes all 14 pre-verifier rules and the 10-trial adversarial verifier (mean blind-solve rate 0.7%), and is R14-discriminating (intended ≠ default answer beyond grader tolerance).

Table 5: FinInteract dataset statistics. Disambiguation entropy $H_0 = \log_2 k$ over k plausible interpretations.

Property	Value
Total instances	173
Language (EN / ZH)	53 / 120
Source (EDGAR / CSRC-CNINFO / DocFinQA)	50 / 120 / 3
Filing type (10-K / 年报)	53 / 120
Unique companies	61
Filing years (2022/23/24/25)	40 / 53 / 58 / 19
Primary axis: entity_scope	94
metric_definition	63
recognition_policy	9
temporal_scope	7
filing_vintage	0
# axes (single / two)	128 / 45
H_0 (mean / median / range)	2.20 / 2.00 / 1.0–3.6
Mean blind-solve rate (verifier)	0.7%

5 EVALUATION FRAMEWORK

5.1 Agent Architecture

We evaluate agents under the ReAct framework in seven configurations (Table 2). The **interact** action presents a yes/no question to a simulated user. The user simulator (GPT-5 at temperature 1.0) is given the disambiguating context C and responds yes, no, or "I don't know." To test robustness, we evaluate under three simulator variants: **LLM** (primary; GPT-5, T = 1.0), **Oracle** (deterministic keyword matching against $\mathcal{I}_{\text{int}}$), and **Noisy** (15% random answer corruption).

5.2 Models

Eight models are evaluated across all three primary modes: GPT-5, GPT-5-mini, GPT-4o (OpenAI); Claude-Sonnet-4, Claude-Opus-4 (Anthropic); DeepSeek-V3.1, Qwen3-235B, GLM-4.5 (open-weight).

5.3 Metrics

We deliberately lead with metrics that are standard in the ambiguous-QA and interactive-retrieval literature (accuracy, calibration, interaction counts), so that FinInteract is directly comparable to prior benchmarks rather than self-referential. A single bespoke efficiency score is reported only as a supplementary diagnostic.

Accuracy (%) (primary) is the fraction of instances whose final answer matches the *intended* interpretation, graded by GPT-4o-mini with financial-domain tolerance (±1% numeric, entity/ticker equivalence, currency normalization; fiscal-year mismatch = wrong). This is the metric on which models are ranked, mirroring InteractComp, BrowseComp, and FinanceBench.

Default-capture rate (%) is the fraction whose final answer instead matches the *default* interpretation. It measures how often a model confidently returns the naive reading, and lets us reconstruct what a conventional single-gold benchmark would have scored (§6.1).

Interaction diagnostics (secondary, descriptive): *IR* (interaction rate, % of instances with ≥ 1 ask), *Round* (mean turns), and *AxisHit*—for each interact action GPT-4o-mini classifies whether the clarifying question targets a true ambiguity axis for that instance; we report AxisHit@1, AnyAxisHit, WrongAxisRate, and GenericAskRate. These characterize *how* a model interacts; they do not rank models.

Calibration: 5-bin Expected Calibration Error (ECE) between stated confidence and correctness, capturing overconfidence on ambiguous queries.

DisE$^+$ (supplementary). As a single number folding accuracy and clarification cost together we report the efficiency of *successful* clarification,

$$\mathrm{DisE}^+ = \mathbf{1}[\text{correct}] \times \mathbf{1}[n_{\text{asks}}{>}0] \times \frac{H_0}{n_{\text{asks}}},$$

which is positive only when the agent asked at least one question and then answered correctly, and decreases with redundant asks. We restrict it to the asked-and-correct case precisely to avoid the perverse incentive of an all-instances variant, which would award its maximum to a confident *zero-ask* answer—rewarding the very behavior the benchmark penalizes. DisE$^+$ is a diagnostic of clarification efficiency, never the ranking metric.

6 EXPERIMENTS

6.1 Why Existing Benchmarks Miss This: The Single-Gold Illusion

Before analyzing model behavior, we establish the core motivation empirically: *a conventional single-gold financial-QA benchmark cannot see the failure FinInteract measures.* Existing benchmarks (FinanceBench, FinQA, DocFinQA) attach one gold answer per question. Built from the same filings, such a benchmark would almost always adopt the *default* reading—the headline figure a non-expert assumes—as that gold. Our paired construction lets us quantify the consequence directly: every instance carries both an intended and a default answer, so we re-grade the *identical* model outputs under both conventions (Table 7).

Finding 0: A single-gold convention over-states competence by up to 3×, and the gap is the benchmark's reason to exist. In the retrieval-augmented regime where current single-gold benchmarks operate, the larger models return the *default* reading more often than the intended one: GPT-4o is graded 37.0% correct against the default but only 12.1% against the intended answer—a single-gold benchmark would report 3.1× its true competence. The model is not reasoning to the right answer; it is confidently producing a well-grounded answer to the *wrong reading of the question*, which a single-gold benchmark rewards and therefore cannot detect. (GPT-5-mini is the instructive exception: it captures *neither* reading because it asks rather than commits (Finding 6), so single-gold grading does not flatter it—but nor does it answer.) Critically, interaction is the *only* mode where intended accuracy exceeds default-capture (GPT-5: 20.2 vs. 12.1): asking is what converts a default-anchored guess into the correct intended answer. This is precisely the capability single-gold benchmarks are structurally blind to, and what FinInteract is built to measure.

6.2 Findings

We report results for the full OpenAI tier ladder—GPT-5, GPT-4o, and GPT-5-mini—across all three modes (Table 6); other-vendor models are in progress. All headline accuracy comparisons below are reported with bootstrap 95% CIs over instances, and we test each within-model mode transition with a two-sided paired bootstrap (n=173). Six findings emerge here, with a seventh (the latent-capacity ceiling) in §6.3.

Finding 1: Interaction is a capability, not a free lever — it helps the strong model and *harms* the weak one. Adding interaction raises GPT-5 from 6.4% to 20.2% (95% CI $[14.5, 26.6]$; paired bootstrap $p<0.001$), but *lowers* GPT-4o from 12.1% to 4.6% (p=0.002). Forcing the weaker model to interact makes it significantly worse than not interacting at all. This extends InteractComp's "interaction stagnation" into a sharper claim: for models that cannot use it, interaction is actively harmful.

Finding 2: Model rankings flip with the interaction axis. GPT-4o *outperforms* GPT-5 under pure retrieval (12.1 vs. 6.4) yet is *dominated* by it under interaction (4.6 vs. 20.2). A single-mode benchmark would mis-rank these models; the interaction axis is what separates them, which is precisely the dimension FinInteract is designed to measure.

Finding 3: Retrieval cannot substitute for interaction. Oracle retrieval lifts GPT-5 to only 6.4%: because each disambiguating passage contains *both* the intended and default values, retrieval surfaces the ambiguity but does not resolve it. Only a targeted clarification collapses the question to a unique answer, which interaction (20.2%) begins to do.

Finding 4: The weak model's failure is clarification non-integration, not inability to ask. GPT-4o asks readily (IR 99%, 4.7 questions on average) but fails to *integrate* the user's responses: 87% of its interactive errors are non-committal refusals (e.g. "please specify the exact period") issued *after* the simulator has answered. It can recognize ambiguity and ask, but cannot close the loop and commit — a multi-turn integration gap, not a protocol artifact (only 2% produced no answer at all).

Finding 5: Models are severely overconfident on ambiguous queries. Expected Calibration Error ranges from 0.53 (GPT-5, interact) to 0.95 (GPT-4o, answer-only); models report 85–95% confidence while answering 0–20% correctly. Both models also ask imprecisely — AxisHit@1 is only 0.38 (GPT-5) and 0.37 (GPT-4o), so even when a model asks, it targets the correct ambiguity axis barely a third of the time, motivating the axis-conditioned policy of §7.

Finding 6: A smaller model collapses into ill-formed clarification — it asks the most but commits the least. GPT-5-mini scores 0.6/0.0/0.0% across the three modes. The cause is not ambiguity-blindness but the opposite: it is the *most* eager to clarify (interact IR 96%, 4.6 asks/instance), yet it issues *open-ended* questions ("which fiscal year? please paste the 2023 and 2022 figures") rather than the protocol's yes/no questions. Manually inspecting all 173 interact trajectories, 63% terminate on an unanswered open-ended question and 35% exhaust the 10-round budget still asking; only 1 instance ever commits a value (also wrong), and just 1.3% were format-unparseable. Even in Answer+Search—where no interact channel exists—it asks instead of answering, capturing *neither* the intended nor the default reading (0.0/0.0%). This is an extreme form of Finding 4's integration gap and a distinct failure mode: clarification without commitment.

Table 6: Main results. Accuracy (%) graded against the intended interpretation is the primary metric (per mode); the interact mode additionally reports the secondary interaction diagnostics IR (% asked), AxisHit@1, and ECE. DisE^{+} and the default-capture comparison are reported separately (Table 7). AxisHit is undefined when no question is asked. The OpenAI tier ladder and two open MoE models (Qwen3-30B-A3B, Qwen3.5-35B-A3B) are evaluated; remaining other-vendor rows in progress. The human baseline is the single interaction condition (passage + one yes/no clarification + answer) on a stratified 50-instance subset; see §6.4. ‡Retrieval-augmented accuracy for the open MoE models is being re-measured with the full passage corpus and is omitted here; their interaction-behaviour columns (IR, AxisHit) are retrieval-independent and reported.

	Acc (%)		Answer+Search+Interact			
Model	Ans-only	+Search	Acc	IR	AHit@1	ECE
GPT-5	0.6	6.4	**20.2**	98.8	.38	.53
GPT-4o	0.6	**12.1**	4.6	99.4	.37	.82
GPT-5-mini	0.6	0.0	0.0	96.0	.34	.00
Claude-Sonnet-4	–	–	–	–	–	–
Claude-Opus-4	–	–	–	–	–	–
DeepSeek-V3.1	–	–	–	–	–	–
Qwen3-235B	–	–	–	–	–	–
GLM-4.5	–	–	–	–	–	–
Qwen3-30B-A3B	1.2	–‡	–‡	72	.23	–
Qwen3.5-35B-A3B	0.0	–‡	–‡	77	.40	–
Proposed method baseline (GPT-5 backbone)						
Axis-Aware ReAct	–	–	–	–	–	–
Human (50-inst)	–	–	**30.0**	100	**.68**	–

Table 7: The single-gold illusion. The same model outputs, graded against the *intended* interpretation (FinInteract) vs. the *default* reading that a conventional single-gold benchmark would treat as gold. "Δ" is the inflation a single-gold benchmark would report. In the retrieval regime where prior benchmarks operate, models capture the default far more than the intended answer; only interaction reverses this.

Model	Mode	Intended (true Acc)	Default (single-gold)	Δ
GPT-5	Answer-only	0.6	2.3	+1.7
	Answer+Search	6.4	10.4	+4.0
	+Interact	**20.2**	12.1	−8.1
GPT-4o	Answer-only	0.6	0.6	0.0
	Answer+Search	12.1	**37.0**	**+24.9**
	+Interact	4.6	6.4	+1.8
GPT-5-mini	Answer-only	0.6	0.0	−0.6
	Answer+Search	0.0	0.0	0.0
	+Interact	0.0	0.0	0.0

Language and axis breakdown. Splitting by language confirms the benchmark's bilingual-difficulty hypothesis: Chinese instances are consistently harder than English in the interact mode (GPT-5 18.3% ZH vs. 24.5% EN; GPT-4o 0.8% vs. 13.2%), reflecting more complex corporate-structure naming and metric conventions in A-share filings. Stratifying by primary axis (interact mode), `metric_definition` is the hardest well-populated axis (GPT-5 12.7%, n=63) and `entity_scope` the next (23.4%, n=94), while `temporal_scope` is comparatively easy (57.1%, n=7). This inverts the general-domain pattern where entity disambiguation dominates, and identifies metric-definition ambiguity as the distinctive financial challenge.

Difficulty and headroom. Per-instance accuracy falls monotonically with disambiguation entropy H_0 (from 10% at H_0=1 to under 1% at H_0≈3.6), validating H_0 as a difficulty proxy; **125 of 173 instances (72%) are solved by no model in any mode**. The best configuration reaches 20.2%, leaving the benchmark far from saturated. The supplementary clarification-efficiency metric mirrors the accuracy ranking in interact mode (DisE^{+} = 0.125 for GPT-5 vs. 0.039 for GPT-4o): GPT-5 not only answers more ambiguous queries correctly but does so with fewer wasted asks.

6.3 Latent Capacity: Forced Interaction vs. the Oracle Ceiling

To locate the bottleneck we bracket interactive accuracy with two ablations (Table 8). **Forced interaction** requires the agent to ask n=4 clarifying questions before answering. **Context-oracle** hands the model the disambiguating context C (which fixes the intended interpretation) together with a passage containing *both* the intended and the default evidence spans, then asks it to answer—the ambiguity is pre-resolved but the model must still ground the correct value.

Finding 7: Models can answer once the ambiguity is resolved—they cannot resolve it themselves. The context-oracle ceiling is 93–95% for all three models, including GPT-5-mini, which scores 0% under free interaction. Latent answering capacity is therefore near-ceiling and roughly scale-invariant; what varies—and what FinInteract measures—is whether a model can *elicit* the disambiguating context. Crucially, *forcing* clarification does not recover this capacity: requiring four asks leaves GPT-4o and GPT-5-mini

Table 8: The disambiguation bottleneck. Forcing more clarification does not help (and often hurts), but *resolving* the ambiguity lifts every model to 93–95%. The ≈75-point gap between the oracle ceiling and free interaction is attributable to the model's inability to elicit C itself—not to missing knowledge or grounding. (†GPT-5 forced capped at n=65. ‡open-MoE interact accuracy pending re-measurement with the full passage corpus; the oracle ceiling injects spans directly and is unaffected.)

Model	Answer-only	Interact	Forced(4)	**Oracle ceiling**
GPT-5	0.6	20.2	1.5†	**95.4**
GPT-4o	0.6	4.6	0.0	**93.1**
GPT-5-mini	0.6	0.0	0.6	**95.4**
Qwen3-30B-A3B	1.2	–‡	–	**90.2**
Qwen3.5-35B-A3B	0.0	–‡	–	**91.9**

near 0% (0.0% and 0.6%) and *lowers* GPT-5 from 20.2% to 1.5%, because forced asks consume the round budget on low-value questions without converging on a committed answer. The bottleneck is targeted, self-initiated disambiguation, which neither scale nor brute-force asking supplies.

6.4 Human Baseline

To bound the task from above under *realistic* interaction—where the disambiguating context must be *elicited*, not handed over—we collect a human baseline on a stratified 50-instance subset (32 EN / 18 ZH, balanced by axis). Two finance-literate annotators (one bilingual) face the same input a model does in the search regime: the ambiguous question and a retrieved passage containing both candidate values, position-shuffled and unlabeled. Each may ask *one* yes/no question, which the experimenter answers truthfully from the withheld context C (Yes/No/IDK), before committing a final answer. This mirrors the agent's `search`→`interact`→`answer` loop, so the human row is directly comparable to the +Interact column; we grade it with the identical GPT-4o-mini grader.

Finding 8: Even humans who can ask resolve only 30%—the task is hard, not merely mis-specified for models. Annotators reach 30.0% accuracy overall (34.4% EN, 22.2% ZH), above every model's ≤ 20.2% but far from the 93–95% context-oracle ceiling. The gap between the oracle ceiling and the human baseline is the cost of *elicitation*: knowing the resolved answer is near-trivial, but extracting the right bit through a single clarification is hard even for experts. Humans ask far more on-target questions than any model (AxisHit 0.68 vs. 0.34–0.38) yet still capture the default reading 50% of the time. The asymmetry is sharpest in Chinese: ZH annotators ask the *best*-targeted questions (0.72 AxisHit) but post the *lowest* accuracy (22.2%) and highest default-capture (72.2%). Chinese instances are therefore harder to *resolve*, not harder to *recognize*—the 归母/扣非 (parent-attributable vs. deduction-of-nonrecurring) net-income distinction is sticky even once correctly surfaced—corroborating the language-asymmetry caveat in §9 with a mechanism.

Table 9: Per-axis Targeting skill (AxisHit@1) in `+interact`. Recognition policy is a near-universal blind spot; temporal scope is largely solved; entity/metric are partial. "–": model never asked on that axis (IR= 0). The two sparse axes (n≤9) are preliminary.

Axis	n	GPT-5	GPT-4o	GPT-5-m	Q3-30B	Q3.5-35B
entity_scope	94	.41	.29	.37	.28	.39
metric_definition	63	.32	.13	.27	.18	.36
recognition_policy	9	**.00**	**.00**	**.00**	**.00**	.17
temporal_scope	7	.95	1.0	.95	–	1.0

6.5 Per-Axis Skills: Which Clarification Abilities Do Models Lack?

The five axes are not just data strata; each names a distinct *skill*—the ability to recognize that a question is under-specified *along that axis* and ask the question that resolves it. We make this measurable by decomposing each model's per-axis `+interact` competence into three components: **Recognition** (interaction rate—does it ask when it should?), **Targeting** (AxisHit@1—does it ask about the *right* axis?), and **Resolution** (accuracy—does asking yield the intended answer?). A model "has" an axis skill only when all three are high; a low value localizes the failure to not-asking, asking-wrong, or asking-right-but-still-wrong. Table 9 reports Targeting, the decisive component.

Finding 9: Ambiguity clarification is a per-axis skill, and models lack specific ones. Targeting is sharply differentiated: every OpenAI model scores AxisHit = 0 on `recognition_policy` (they essentially never ask whether a figure is recognized point-in-time vs. over time), yet all reach ≥ 0.95 on `temporal_scope`; entity and metric sit in between (0.13–0.41). Resolution follows the same shape (e.g. GPT-5 resolves 57% of temporal but ≤ 23% of entity/metric). The gap is therefore not uniform incompetence but *missing specific skills*—which is precisely the target an intervention should aim at (§7.4). We also tested whether failures cluster by axis in per-instance behavior space (PCA over per-model correct/ask/AxisHit/default); they do not (k-means ARI ≈ 0.01, p ≈ 0.15), a floor effect of the near-zero accuracies—the axis structure lives in the aggregate targeting profile, not in coarse per-instance behavior. Clustering the *text of the asked clarifying questions* is the sharper test and is left to future work.

6.6 Mechanistic Case Study: Where Ambiguity Lives in the Network

The behavioral results (§6) show that frontier models default to the naive interpretation rather than clarifying. A natural question is whether models internally *represent* this ambiguity at all. Because this requires access to hidden states, we probe four open models as a proxy, spanning two size classes and two architectures: **Qwen3-4B-Instruct** (dense, 37 layers), **Qwen3.5-4B** (hybrid linear/full attention, 33 layers), and the mixture-of-experts **Qwen3-30B-A3B** (49 layers) and **Qwen3.5-35B-A3B** (41 layers).

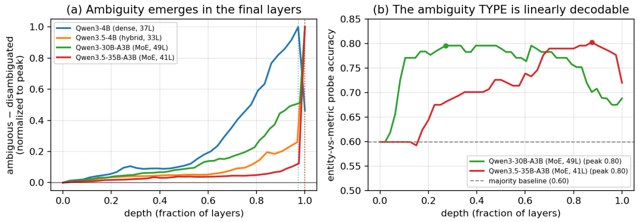

Figure 1: Four-model depth profile of the ambiguity signal (Qwen3 4B→35B, dense and MoE). (a) Ambiguous−disambiguated separation rises to a peak in the final layers of every model (dotted verticals mark per-model peaks; normalized for depth and magnitude). (b) The ambiguity *type* (entity vs. metric) is linearly decodable at 0.80 on both MoE models, well above the 0.60 majority baseline, peaking at different depths.

Method. For each instance we form a within-item contrast: the bare ambiguous question Q versus Q concatenated with its disambiguating context C. We define a per-layer *ambiguity direction* as the difference in mean final-token activations between the two conditions, and score any hidden state by its (mean-centered) projection onto that direction; tracing this across layers gives a depth profile of how strongly each layer separates an under-specified question from a disambiguated one. Independently, on the two larger models we train a per-layer linear probe to decode the ambiguity *type* (entity_scope vs. metric_definition), which—unlike a context-presence detector—cannot be solved trivially.

Result. Figure 1(a) overlays the depth profiles (normalized for differing depth and scale). On all four models the signal is near-zero through most of the stack and rises sharply to a peak in the *final few layers* (≈95–98% relative depth: layers 35/37, 32/33, 48/49, 40/41). Figure 1(b) shows that the ambiguity *type* is linearly decodable at 0.80 accuracy (majority baseline 0.60) on both larger models, though it becomes available at different depths—early (≈27%) in Qwen3-30B and late (≈85%) in Qwen3.5-35B. Across four models, two scales, and three architecture families, the representation is consistent: financial-query ambiguity, and its axis, is linearly encoded, emerging toward the top of the network.

Representation and elicitation. The two MoE models we probe are also *evaluated* (Table 6): beyond linearly encoding the axis, both *elicit* —interaction rates of 72–77% with on-axis targeting (AxisHit 0.40 vs. 0.23)—and differ in *how* they ask, Qwen3-30B looping without committing in Chinese (76% of its interactive transcripts never produce a final answer) where Qwen3.5-35B commits in 95% of cases. These interaction-behaviour signals (rate, axis targeting, commitment) are independent of retrieval; the models' retrieval-augmented *accuracy* is being re-measured with the full passage corpus and is therefore omitted here. The representational and behavioural evidence shows the axis is both registered and acted on at the level of asking. A linear-probe replication on the *base* (non-instruct) checkpoints reproduces the decodability and the early-vs-late depth split (peaks 0.71/0.74 vs. baseline 0.60).

Finding 10: Open models linearly represent both ambiguity and its axis, replicating from 4B to 35B and across dense and MoE architectures. The representation is type-specific (entity vs. metric decodable at 0.80) and robust to scale and architecture, yet behaviorally these same models still default rather than clarify. The behavioral gap is thus better read as a failure to *act* on a represented ambiguity than a failure to register it—though establishing this causally (e.g. via activation steering) remains future work.

Scope and caveats. This is mechanistic evidence on open proxies, not the closed frontier models benchmarked in §6; diff-in-means probing and linear decoding establish *availability*, not causal use; and axis-type decoding is supported only for the two well-populated axes (entity_scope, metric_definition). We deliberately do *not* draw behavioral conclusions from these open models' free-form generations, where surface cues (chain-of-thought markers, retrieval that surfaces the evidence span) confound naive clarification counts; the asking-behavior claims come solely from the graded closed-model evaluation in §6. That the axis is *represented but not acted on* frames the bottleneck as an action problem on an available signal—and motivates the axis-conditioned interventions of §7, which target exactly that signal.

7 DIAGNOSIS AND INTERVENTION: ACTING ON THE AMBIGUITY AXIS

Our results converge on a single diagnosis. The benchmark localizes the bottleneck to *eliciting* the disambiguation: agents answer 93–95% correctly once the intended interpretation is supplied (§6.3) but resolve only ~8% on their own, and the mechanistic probe shows why—open models *linearly represent* the ambiguity *axis* yet do not act on it (§6.6). The remaining gap is thus an *action* problem on an already-available signal, and the *axis* is its natural unit. This section turns that diagnosis into intervention: we first catalog how acting fails (a seven-type error taxonomy), then present two interventions that share one idea—**condition on the axis**—instantiated at inference time (Axis-Aware ReAct, §7.3) and at training time (axis-guided SFT/GRPO, §7.4).

7.1 Error Taxonomy

Beyond aggregate accuracy, we characterize model failures using a seven-type error taxonomy (Table 10) derived from FinInteract's evaluation signals. Each type is diagnosable from existing metrics without additional annotation.

7.2 Diagnostic Mapping from Evaluation Signals

The existing evaluation metrics map directly onto error types, enabling failure attribution without additional annotation:

- **Low IR + low Acc** → E1 (ambiguity blindness dominant)
- **High IR + low AxisHit@1** → E2/E3 (interacts but misdirected)
- **High AxisHit@1 + low Acc** → E5/E6 (right axis, wrong grounding)
- **High GenericAskRate** → E3 (lacks financial clarification structure)
- **High WrongAxisRate** → E2 (confidently misdirected)

Table 10: Error taxonomy E1–E7 with diagnostic signals and improvement directions.

Type	Description	Improvement direction
E1	**Ambiguity blindness**: no interaction, wrong answer	Ambiguity pre-check; train on detection
E2	**Wrong-axis clarification**: asked wrong dimension	Axis classifier; financial taxonomy grounding
E3	**Generic clarification**: vague ask, no axis target	Axis-conditioned clarification templates
E4	**Retrieval dominance**: searched but skipped interaction	Force interpretation check after retrieval
E5	**Clarification misuse**: right question, wrong answer	State-gated answering; stronger grounding
E6	**Evidence grounding**: template-oracle correct, agent wrong	Improved source-grounded extraction
E7	**Over-interaction**: correct but low DisE$^+$	Penalize via DisE$^+$/AxisHit efficiency signal

- **Always-ask** $\gg$ **Standard ReAct** → E1 dominant; latent capacity exists
- **Template-oracle** $\gg$ **Agent-interact** → "what to ask" is the bottleneck
- **Axis-oracle** $\gg$ **Agent-interact** → axis recognition is the bottleneck
- **Template-oracle still low** → E6 residual; evidence extraction still fails

7.3 Inference-Time Intervention: Axis-Aware Clarification ReAct

The first intervention conditions on the axis *without changing the model weights.* Standard ReAct leaves the agent free to decide whether and what to ask. We propose **Axis-Aware Clarification ReAct**, a three-component structured interaction policy that directly addresses E1–E3.

Component 1: Ambiguity Pre-Check. Before any action, the agent explicitly decides which of the five ambiguity axes are underspecified in the question and initializes an interpretation state $\mathcal{S}$ = {entity, period, metric, basis, vintage}, marking each field as *resolved* or *unknown.*

Component 2: Axis-Conditioned Clarification. For each *unknown* field, the agent generates a targeted yes/no question using axis-conditioned templates (one per axis) rather than free-form clarification. The agent emits at most one clarification per turn, prioritizing the most information-theoretically valuable unknown field.

Component 3: Interpretation-State-Gated Answering. The agent may only emit a final answer once all required fields are resolved (or the user explicitly declines to clarify). This prevents the *retrieval dominance* pattern (E4) where the agent answers immediately after finding a plausible number in a search result.

Table 11: RLVR ladder on FinInteract (Qwen3-4B, held-out n=51). AxisHit@1 and interaction are leak-proof signals of learned clarification; accuracy (*) is partly leaked by oracle retrieval and should not be read as a learning signal.

Stage	AxisHit@1	Interaction	Accuracy*	Reward
Base (Qwen3-4B)	25.0	54.9	84.3	0.96
+ SFT (axis-guided)	68.6	**100**	88.2	1.28
+ KTO	**70.6**	**100**	84.3	1.27
+ GRPO	66.7	**100**	88.2	**1.30**

Expected outcomes. Relative to standard Answer+Search+Interact, Axis-Aware ReAct should reduce GenericAskRate (E3), WrongAxisRate (E2), and IR=0 failures (E1), while improving AxisHit@1 and ultimately Accuracy. Template-oracle provides an upper bound: if Axis-Aware ReAct approaches template-oracle performance, it confirms that axis recognition is the dominant remaining bottleneck after the E1–E3 errors are addressed.

7.4 Training-Time Intervention: Axis-Guided SFT and GRPO

The per-axis skill analysis (§6.5) gives this intervention a precise target: models do not lack clarification ability uniformly—they lack *specific* axis skills (AxisHit = 0 on recognition policy, partial on metric). The question is whether those missing skills can be installed by training rather than prompting. The second intervention applies the *same* idea—condition on the axis—but in the weights rather than the prompt, showing the benchmark is not only diagnosable but *trainable.* Starting from Qwen3-4B-Instruct (4-bit QLoRA), we run an RLVR ladder—supervised fine-tuning (SFT), KTO, then GRPO—on the FinInteract training split, with all user simulators, graders, and axis judges served locally (zero API cost). We score each stage with the two *leak-proof* metrics, AxisHit@1 and interaction rate (Table 11); accuracy is reported only with the caveat that the training environment's retrieval returns the disambiguated evidence span, so it is partly leaked and is *not* a reliable learning signal.

The axis-guided teacher. The decisive ingredient is how SFT demonstrations are generated. An uninformed teacher—even a 72B model—almost always asks the *obvious* ambiguity (typically the reporting period) and rarely the subtle gold axis (entity scope, recognition policy), yielding ~0% on-axis demonstrations. Privately revealing the gold axis to the teacher while generating demonstrations (the hint is *not* stored in the trajectory) lifts the on-axis yield to ~79%—a reusable recipe for any axis-structured clarification task.

SFT does the heavy lifting. The leak-proof signals jump almost entirely at the supervised stage—AxisHit@1 rises 25 → 69 and interaction 55 → 100 (Figure 2a)—while KTO and GRPO move within run-to-run noise at n=51. Training teaches the model to *always interact and target the correct axis*, the behavior FinInteract is built to elicit. Whether this gain reflects a sharpened internal *representation* of the axis or only a rewired output *policy*—probing the trained model as in §6.6—is a direct test of the diagnosis and a target for future work.

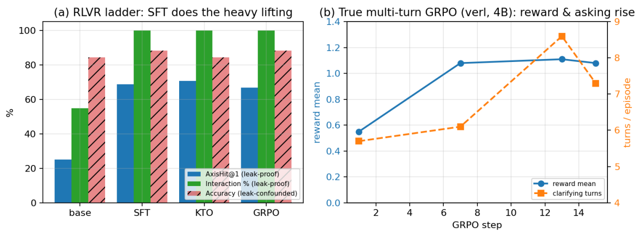

Figure 2: Training on FinInteract. (a) The leak-proof learning signals (AxisHit@1, interaction) jump at SFT and saturate; accuracy (hatched) is leak-confounded. (b) True multi-turn GRPO on the 4B policy: mean reward and clarifying turns rise together over training.

True multi-turn GRPO. The ladder above optimizes a single-turn approximation. We additionally confirm the full multi-turn episode is trainable end-to-end: optimizing the 4B policy over entire ReAct episodes (loss masked on simulator and retrieval tokens) for 15 steps raises mean reward 0.55 $\longrightarrow$ 1.08 while clarifying turns rise 5.7 $\longrightarrow$ 8.6 (Figure 2b)—the axis-aware reward is teaching the policy to ask more. We report this as a learning-signal demonstration, not a converged model.

Caveats. Results are on a held-out split of n=51; all judges and the teacher are local Qwen2.5-32B/72B models (weaker than GPT-4o, and siblings of the policy); the terminal reward is partly leaked; and GRPO is run for tens of steps, not to convergence. We therefore frame this as evidence that FinInteract is *trainable*—an axis-guided SFT seed instills axis-targeted clarification—rather than as a state-of-the-art accuracy result.

8 RELATED WORK

Financial QA Benchmarks. Table 12 compares FinInteract to major financial QA benchmarks. FinanceBench [4], FinQA [2], DocFinQA [10], BizBench [5], FinBen [13], and FinanceQA [8] all enforce single-gold-answer conventions and evaluate models in a single turn; FinanceQA further shows that frontier models fail ~60% of realistic analyst tasks, but attributes this to numerical rigor rather than ambiguity. FinEval [3], a large Chinese financial-knowledge benchmark, incidentally identifies an "ambiguity handling weakness" error type in its agent tasks but does not isolate or control it. FinInteract is the first to make query ambiguity the central, controlled construct—via paired default/intended interpretations and multi-turn interaction—in the financial domain.

Ambiguous QA.. AmbigQA [9], ASQA [11], CondAmbigQA [7], and CLAM [6] have matured in the general (Wikipedia) domain. We adapt their formalism to finance, replacing human-judged ambiguity labels with programmatically verifiable XBRL-grounded evidence pairs.

Interactive Search Agents. InteractComp [1] is the direct methodological ancestor. We extend its general-domain setting to finance with a domain-specific ambiguity taxonomy and bilingual evaluation. BrowseComp [12] tracks longitudinal retrieval trends; we test whether the interaction-capability gap identified there persists in finance.

9 LIMITATIONS

Scale. At 173 instances FinInteract is small (comparable to FinanceBench's 150 [4]); we prioritize per-instance construction rigor (every item is adversarially verified and R14-discriminating) over raw size, and Appendix A shows the pipeline scales at \$0.31/instance. **Language balance.** The English split (53) is smaller than the Chinese split (120); per-language results (§6) should be read with this asymmetry in mind, and the EN portion is a target for growth. **Axis coverage.** The taxonomy is intentionally five-axis, but `filing_vintage` is unrepresented and `temporal_scope`/`recognition_policy` are sparse (Table 5): viable structured ambiguity for these axes is rare in public filings (Appendix A). **Model coverage.** We report the OpenAI tier ladder (3 models); other-vendor models are in progress. The human baseline (N=50) establishes the performance ceiling but is collected from two annotators on a subset; a larger expert panel remains future work. **Grader and simulator.** Correctness and axis-targeting are judged by GPT-4o-mini and the user is simulated by GPT-5; although 1.3% of interact outputs were format-unparseable and manually checked, a full judge-vs-human agreement study remains future work. **Contamination.** Instances are freshly constructed from FY2022–2025 structured facts and answer-only accuracy is ≈0.6% across all sources, but we do not claim absolute contamination safety for the DocFinQA-derived minority.

10 CONCLUSION

We introduced FinInteract, the first bilingual benchmark for evaluating financial search agents on genuinely ambiguous queries. The five-axis financial ambiguity taxonomy, paired default/intended interpretation design, and adversarial verifier-based construction pipeline yield 173 high-quality instances that resist trivial disambiguation. One unit organizes the whole story—the ambiguity *axis*: it structures the taxonomy, is what an agent must elicit (single-gold grading overstates competence up to 3× when models commit to the wrong axis-reading), is *linearly represented* inside open models yet left unacted-on, and is the lever for intervention. Our central finding is that interaction capability—not financial knowledge—is the bottleneck: agents answer 93–95% correctly once the axis is resolved but ~8% on their own, amplifying the stagnation InteractComp identified in the general domain. We turn this diagnosis into intervention with two axis-conditioned methods—at inference (Axis-Aware ReAct) and at training (axis-guided SFT/GRPO)—and release FinInteract with full construction, evaluation, and training code for reproducible follow-on work.

Table 12: Comparison of financial QA benchmarks. ✓ = supported; – = not applicable or not evaluated.

Benchmark	Scale	Ambiguity	Interaction	Multi-turn	Bilingual	Evidence	Grounding
FinanceBench [4]	150	–	–	–	–	human	SEC
FinQA [2]	8,281	–	–	–	–	human	10-K/Q
DocFinQA [10]	7,437	–	–	–	–	human	10-K/Q
BizBench [5]	multi-task	–	–	–	–	auto	mixed
FinBen [13]	multi-task	partial	–	–	partial	auto	mixed
FinanceQA [8]	multi-task	–	–	–	–	expert	SEC
FinEval [3]	8,351	–	–	–	ZH	expert	exams
FinInteract (ours)	173	✓	✓	✓	EN+ZH	LLM+human	EDGAR+CNINFO

A CURATION COST TO REPLICATE THE BENCHMARK

A practical concern for any LLM-assisted benchmark is the cost to reproduce or scale it. We instrument the construction pipeline (§4) and report measured API costs. All figures use list prices at construction time (per 1M tokens: GPT-5 $1.25 in / $10.00 out; GPT-5-mini $0.25 / $2.00; GPT-4o-mini $0.15 / $0.60) and count API fees only—they exclude the one-time filing-collection compute and the human spot-check labor (10–20% of accepted instances).

The constructor, not the verifier, is the yield bottleneck. Re-running the pipeline over a held-out sample of candidate passages reproduces a sharp funnel (Table 13). Only 11.5% of keyword-tagged candidate passages survive the constructor, which rejects the rest as containing *no viable ambiguity* for the target axis; but of those that survive, 98.5% pass the 10-trial adversarial verifier. In other words, the construction prompt is strict and the verifier rarely needs to fire—the cost of the benchmark is dominated by constructor calls spent on candidate passages that turn out not to support a genuine ambiguity, not by verification.

Table 13: Measured curation cost. Unit costs are from instrumented pipeline runs (constructor-only n=20; full pipeline n=6); the acceptance funnel is from the full construction run (n=1127 candidate passages). A full pipeline pass is one GPT-5 constructor call plus ten verifier trials (five GPT-5, five GPT-5-mini) with GPT-4o-mini grading.

Quantity	Value
Constructor viability (viable / candidate)	11.5%
Verifier acceptance (accepted / viable)	98.5%
Overall acceptance (accepted / candidate)	11.3%
Cost per rejected candidate (constructor only)	$0.020
Cost per viable passage (full pipeline)	$0.155
Blended cost per candidate passage	$0.035
Cost per accepted instance	**$0.31**

Projected cost to scale. At the measured blended rate, producing N accepted instances requires $\approx N/0.113$ candidate passages and the following API spend: 173 instances (the current release) ≈ $54; 500 instances ≈ $155; 1000 instances ≈ $309. The benchmark is therefore inexpensive to scale in API terms—an order of $300 takes it to 1000 instances—and the true constraint is the supply of candidate passages that contain structured, verifiable ambiguity. This favors *better passage pre-selection* (raising the 11.5% viability rate) over brute-force compute, and explains why the rare axes (e.g. filing_vintage), whose candidate passages almost never survive the constructor, are the expensive ones to grow.

B REPRODUCIBILITY

Models and decoding. Closed models are accessed via their official APIs (evaluated 2026): `gpt-5`, `gpt-4o`, `gpt-5-mini`. Reasoning models (gpt-5 family) use default temperature with a generous `max_completion_tokens` headroom (+4096 above the output budget) so internal reasoning does not truncate the answer; other models use temperature 0 for the agent and grader. **User simulator.** GPT-5 at temperature 1.0, constrained to {Yes, No, I don't know} and conditioned on the disambiguating context C. **Grader.** GPT-4o-mini at temperature 0, binary correctness with finance tolerance (±1% numeric, entity/ticker equivalence, currency normalization; fiscal-year mismatch = wrong). **Protocol.** ReAct with `search/interact/answ` actions, max 10 rounds; the forced-interaction ablation requires n=4 asks before answering; the context-oracle ceiling supplies C plus both evidence spans (position-shuffled). **Statistics.** Accuracy CIs are 95% bootstrap over instances (B=2000); mode transitions use a two-sided paired bootstrap (B=5000). All evaluation, construction, and analysis scripts are released.

C DATASHEET AND LICENSING

Provenance. English instances derive from public SEC EDGAR filings (10-K, FY2022–2025) and a small DocFinQA-sourced set; Chinese instances derive from public CSRC annual reports (年报) retrieved via CNINFO/akshare. All source documents are public regulatory filings; no proprietary, paywalled, or personal data is used. **Schema.** Each instance is a JSON record with fields: `instance_id`, `language`, `source`, `ticker`, `company`, `filing_type`, `filing_date`, `question`, `context` (C), `answer`, `default_answer`, `intended_evidence_span`, `default_evidence_span`, `intended_interpretation/default_interpretati` ({entity, period, metric, basis}), `axes`, `n_axes`, `h0`, and `qc` (per-rule pass flags + verifier blind-solve rate). **Intended use.** Evaluating whether interactive search agents recognize and resolve query ambiguity; not a training corpus and not financial advice. **License and release.** The benchmark, construction pipeline, and evaluation code are released for research use; redistribution of source filings follows the respective public-domain (EDGAR) and regulatory-disclosure (CSRC) terms. **Maintenance.** The frozen v1 release is versioned; corrections and the planned EN-split expansion will be issued as numbered releases.

REFERENCES

[1] Anonymous. 2026. InteractComp: Benchmarking Interactive Search Agents on Ambiguous Queries. In *Proceedings of ICLR*. Under review.

[2] Zhiyu Chen, Wenhu Chen, Charese Smiley, Sameena Shah, Iana Borova, Dylan Langdon, Reema Moussa, Matt Beane, Ting-Hao Huang, Bryan Routledge, and William Yang Wang. 2021. FinQA: A Dataset of Numerical Reasoning over Financial Data. In *Proceedings of EMNLP*.

[3] Xin Guo, Haotian Xia, Zhaowei Liu, Hanyang Cao, Zhi Yang, Zhiqiang Liu, Sizhe Wang, Jinyi Niu, Chuqi Wang, Yanhui Wang, et al. 2023. FinEval: A Chinese Financial Domain Knowledge Evaluation Benchmark for Large Language Models. *arXiv preprint arXiv:2308.09975* (2023).

[4] Pranab Islam, Anand Kannappan, Douwe Kiela, Ruby Qian, Nino Scherrer, and Bertie Vidgen. 2023. FinanceBench: A New Benchmark for Financial Question Answering. In *arXiv:2311.11944*.

[5] Michael Krumdick, Jenna Lorber, Christopher Tsui, Varshini Vithala, Mihir Bhatt, Fei Kuo, Harish Tayyar Madabushi, and Rajiv Mazumder. 2024. BizBench: A Quantitative Reasoning Benchmark for Business and Finance. In *Proceedings of ACL*.

[6] Lorenz Kuhn, Yarin Gal, and Sebastian Farquhar. 2022. CLAM: Selective Clarification for Ambiguous Questions. In *ICML Workshop on Knowledge Retrieval and Language Models*.

[7] Ziyun Li et al. 2025. CondAmbigQA: A Benchmark for Conditional Ambiguous Question Answering. In *Proceedings of EMNLP*.

[8] Spencer Mateega, Carlos Georgescu, and Danny Tang. 2025. FinanceQA: A Benchmark for Evaluating Financial Analysis Capabilities of Large Language Models. *arXiv preprint arXiv:2501.18062* (2025).

[9] Sewon Min, Julian Michael, Hannaneh Hajishirzi, and Luke Zettlemoyer. 2020. AmbigQA: Answering Ambiguous Open-domain Questions. In *Proceedings of EMNLP*.

[10] Varshini Reddy, Rik Balaguer, Ganesh Bhatt, Mihir Bhatt, Bowen Jiang, Fei Kuo, Ching-Yun Lee, Harish Tayyar Madabushi, Rajiv Mazumder, Nandana Mihindukulasooriya, Pedro Morales, and Guilherme Picco. 2024. DocFinQA: A Long-Context Financial Reasoning Dataset. In *Proceedings of ACL*.

[11] Ivan Stelmakh, Yi Liang, Nouha Dziri, Nanyun Peng, and Kai-Wei Chang. 2022. ASQA: Factoid Questions Meet Long-Form Answers. In *Proceedings of EMNLP*.

[12] Jason Wei et al. 2025. BrowseComp: A Simple yet Challenging Benchmark for Browsing Agents. *arXiv:2504.12516* (2025).

[13] Qianqian Xie, Weiguang Han, Zhengyu Zhang, Ruoyu Lai, Min Peng, Alejandro Lopez-Lira, and Jimin Huang. 2024. FinBen: A Holistic Financial Benchmark for Large Language Models. In *Proceedings of NeurIPS Datasets and Benchmarks Track*.

[14] Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik Narasimhan, and Yuan Cao. 2023. ReAct: Synergizing Reasoning and Acting in Language Models. In *Proceedings of ICLR*.